The Canary in the Mine: Game Theory as Diagnostic Instrument for the Erosion of Human Agency

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Abstract

Game theory is typically treated as a predictive tool for strategic interaction. When its predictions fail, we assume model misspecification and seek refinements. I propose a different interpretation: game theory's foundational assumptions formalize the structural conditions of human agency, and their erosion explains why the models fail. The four core assumptions, separate players with distinct preferences, imperfect information, bounded rationality, and minimal symmetry in affecting outcomes, correspond to conditions that must hold for individuals to be strategic agents rather than administered objects. When these conditions collapse, interaction transforms from game to administration; the weaker party ceases to be a player and becomes a variable in another's optimization. I develop this framework through three analytical moves. First, I reread the Hayek-Mises tradition in economics as a defense of agency conditions rather than merely efficiency. Second, I present a thought experiment involving extreme information asymmetry to reveal the structure of complete condition-collapse. Third, I apply Richard Dawkins's extended phenotype framework to explain how administrative systems emerge: ideologies that attack agency conditions can be understood as memetic replicators constructing institutional machinery for their propagation. The framework has implications for AI governance (preserving agency may require limiting information asymmetry regardless of intentions), institutional design (privacy, pluralism, and antitrust are structural conditions of agency, not merely preferences), and political philosophy (freedom requires genuine uncertainty, not merely absence of constraint). Game theory functions as a canary in the coal mine: when it stops describing, when its models systematically fail, the question is not how to improve the models but what has happened to the conditions of human agency.

Keywords: game theory, human agency, information asymmetry, extended phenotype, Hayek, institutional design, AI governance

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I. Introduction: When the Canary Stops Singing

Coal miners once carried canaries into the tunnels. The birds, more sensitive to toxic gases than humans, served as early warning systems. When the canary stopped singing, miners knew to evacuate. The canary's silence was not a defect in the canary; it was information about the environment.

Game theory, I argue, functions as a canary for human agency.

The standard view treats game theory as a predictive tool. It models strategic interaction: situations where multiple agents make choices that affect each other's outcomes. When game-theoretic predictions fail, we typically conclude that our model was misspecified, our assumptions unrealistic, our analysis incomplete. We try to build better models.

I propose a different interpretation. Game theory's foundational assumptions are not arbitrary simplifications for mathematical convenience. They formalize the structural conditions under which human beings can be strategic agents rather than administered objects. When those conditions erode, game theory fails to describe. This failure is diagnostic. It tells us something has changed in the environment.

The four core assumptions are:

1. Separate players: Distinct individuals with non-identical preferences
2. Imperfect information: Limits to what each player knows about others
3. Bounded rationality: Cognitive constraints that make behavior imperfectly predictable
4. Minimal symmetry: Some capacity of each player to affect outcomes

These are typically treated as modeling conveniences. I treat them as conditions of agency. When all four hold, interaction is genuinely strategic: agents pursue their ends through choices that matter. When they erode, interaction transforms into something else. The weaker party ceases to be an agent and becomes a variable in another's optimization.

The argument proceeds as follows. Section II examines each assumption in detail, showing what it formalizes and what attacks it. Section III connects this framework to the Hayek-Mises tradition in economics, rereading their defense of dispersed knowledge as a defense of agency conditions. Section IV presents a thought experiment that collapses all four conditions simultaneously, revealing the structure through an extreme case. Section V applies the extended phenotype framework to explain how administrative systems emerge and persist: ideologies that attack agency conditions may be understood as memetic replicators building institutional machinery for their propagation. Section VI applies the diagnostic to historical and contemporary cases. Section VII explores implications for AI governance, institutional design, and political philosophy. Section VIII addresses limitations and objections.

The central claim is simple: game theory is a canary. When it stops singing, when its models fail to describe, when its predictions systematically miss, we should not conclude we need better game theory. We should ask what has happened to the conditions of human agency.

II. The Four Conditions of Strategic Agency

Game theory is typically presented as a mathematical framework for analyzing strategic interaction. Its foundational assumptions, rarely examined in philosophical terms, are treated as technical conveniences that simplify modeling. I propose a different reading: these assumptions formalize the structural conditions under which individuals can be strategic agents rather than administered objects. When we examine what game theory requires to function, we discover what human agency requires to exist.

II.A. Separation: The Possibility of Distinct Ends

Standard game theory assumes multiple players with non-identical preference orderings. Player 1's utility function differs from Player 2's; otherwise, there would be no game, only a coordination exercise with a single collective preference. This assumption is not merely mathematical. It formalizes something deeper: the existence of individuals with distinct identities, private spheres, and diverse ends.

The Prisoner's Dilemma illustrates the point. Two suspects, interrogated separately, must decide whether to betray the other without knowing what their partner will choose. The dilemma's structure presupposes genuine separateness: distinct minds that cannot directly access each other's intentions, distinct interests that may diverge, distinct decision processes occurring in isolation. Remove the separation, and the dilemma dissolves. If the prisoners shared a single mind, they would simply coordinate on the optimal joint outcome. The "problem" of cooperation exists only because the players are genuinely separate.

This has implications beyond game theory. Coordination problems, trust deficits, negotiation difficulties: all presuppose that agents are not merely nodes in a unified system. When theorists speak of "solving" coordination problems through institutional design, they often imagine eliminating the friction that separation creates. But that friction is not a bug. It is the space within which individual agency operates.

What attacks this condition:

Any project that seeks to homogenize preferences, eliminate private spheres, or absorb individuals into collective identities directly attacks the first condition. Mandatory ideological conformity, surveillance of private thought, and "coordination" that means absorption rather than negotiation all erode the separation that makes strategic interaction possible.

Diagnostic signal:

When "coordination games" have only one acceptable outcome, when "negotiations" are rituals confirming predetermined results, when "diverse perspectives" must converge on a single approved position, the condition of separation has been compromised. The game-theoretic model will fail to describe what is happening, not because the model is wrong, but because the situation is no longer a game.

II.B. Imperfect Information: The Space for Strategy

Much of modern game theory concerns games of incomplete or imperfect information. Bayesian games model situations where players are uncertain about each other's "types," their preferences, their constraints, their private information. This uncertainty is not a complication to be eliminated; it is the source of strategic depth.

Consider poker. The game exists because players cannot see each other's cards. Every bet, every raise, every fold is a signal that might be true or might be bluff. Players must infer from observable actions what hidden realities might be. Remove the hidden information, and poker becomes a trivial calculation. There is no strategy because there is nothing to strategize about.

The same logic applies beyond card games. In any negotiation, each party knows something the other does not: their true reservation price, their outside options, their time constraints, their preferences across different deal structures. This mutual uncertainty creates the space for bargaining, for creative deal-making, for outcomes that neither party could have predicted in advance. If one party knew everything the other knew, "negotiation" would become theater, a performance whose outcome was already determined.

Hayek understood this in economic terms. His argument for markets rested partly on the impossibility of centralizing dispersed knowledge. But the point extends beyond epistemology. Even if knowledge could be centralized, doing so would transform interaction from strategic engagement to unilateral optimization. The other party would become not a player to negotiate with, but a system to be modeled and manipulated.

What attacks this condition:

Surveillance apparatus, mandatory disclosure requirements, data aggregation that allows one party to model another with high accuracy, elimination of privacy: all attack the condition of imperfect information. When one actor can predict another's behavior with near-certainty, the predicted party is no longer a strategic agent. They are a variable.

Diagnostic signal:

When one party can anticipate all moves before they occur, when "surprises" never happen, when strategic innovation disappears and behavior becomes formulaic, the information condition has collapsed. Models that assume genuine uncertainty will systematically mispredict, not because they are poorly specified, but because uncertainty has been engineered out of the system.

II.C. Bounded Rationality: The Reality of Cognitive Limits

Herbert Simon won the Nobel Prize in Economics partly for demonstrating that humans do not optimize; they satisfice. Faced with complex decisions, people use heuristics, rules of thumb, pattern recognition, intuition. They cannot calculate all possible futures; they simplify, approximate, and often err.

Game theory traditionally assumed perfect rationality, the ability to compute optimal strategies given payoff structures. But even in games of complete information, genuine optimization is often computationally infeasible. Chess has more possible positions than atoms in the observable universe. No human, and arguably no computer, can calculate all consequences of all moves. Chess masters play well not because they compute everything, but because they recognize patterns, evaluate positions intuitively, and make educated guesses.

This limitation is not merely practical; it is constitutive of what makes games interesting. If both players could calculate optimal strategies, games would be solved. Outcomes would be determined by initial positions, not by skill or creativity or luck. Tic-tac-toe is boring because its small state space allows complete analysis. Chess remains interesting because its complexity exceeds analytical capacity.

Bounded rationality creates space for surprise, for learning, for genuine discovery within the game. A player with cognitive limits can be outmaneuvered, can fail to see a threat, can miss an opportunity. This means the other player has something to do: to find the weakness, to exploit the oversight, to create confusion. Strategy exists because perfect calculation does not.

What attacks this condition:

Propaganda that narrows cognitive frames, education systems that produce uniform thinking, algorithmic prediction of behavior that renders individuals transparent, pharmacological or technological interventions that constrain decision patterns: all attack the diversity and unpredictability that bounded rationality produces. When every individual processes information identically, when behavior becomes fully predictable, agents have become automata.

Diagnostic signal:

When behavior across a population becomes highly uniform, when responses to stimuli are predictable with mechanical accuracy, when "irrationality" disappears and everyone makes identical choices, the cognitive diversity that bounded rationality describes has been eliminated. Game-theoretic models assuming diverse strategies will fail; not because agents are now "more rational," but because they are no longer genuinely choosing.

II.D. Minimal Symmetry: The Capacity to Affect Outcomes

Game theory does not require equality between players. Many games model asymmetric situations: employer and employee, monopolist and consumer, government and citizen. But even asymmetric games require what might be called minimal symmetry: some capacity of each player to affect the outcome. If one player's actions have no impact on results, they are not playing. They are being done to.

The concept of Nash equilibrium illustrates the point. An equilibrium exists when no player can improve their outcome by unilaterally changing strategy. The definition presupposes that players have strategies that matter, that their choices influence outcomes. If one player's decisions have no effect, the concept of equilibrium collapses into something simpler: the dominant player's optimization, given a fixed environment.

Consider a "negotiation" where one party has complete exit control, unlimited resources, and no time pressure, while the other party has no alternatives, depleted resources, and urgent needs. Formally, this might be modeled as a game. But the outcome is not really negotiated; it is imposed. The weaker party's "strategy" reduces to acceptance or destruction. This is not strategic interaction; it is ultimatum delivery.

What attacks this condition:

Monopolization of alternatives, creation of total dependency, elimination of exit options, vast concentrations of power that make resistance futile: all attack the minimal symmetry that makes games games. When one actor can determine outcomes unilaterally regardless of what others do, those others are not players. They are subjects.

Diagnostic signal:

When "equilibria" are predictable entirely from one party's preferences, when the weaker party's optimal strategy is always compliance, when negotiated outcomes match imposed outcomes, symmetry has collapsed. Game-theoretic analysis may continue to be applied, but it describes theater, not interaction.

II.E. Summary: Four Conditions, One Diagnostic

The four assumptions of game theory are not arbitrary. They formalize what it means for multiple agents to interact strategically:

Assumption	What It Formalizes	Condition of Agency
Separate players	Distinct identities and ends	Individual selfhood
Imperfect information	Limits to mutual knowledge	Epistemic autonomy
Bounded rationality	Cognitive diversity and limits	Genuine unpredictability
Minimal symmetry	Capacity to affect outcomes	Power to resist or exit

When these conditions hold, game theory describes. When they erode, game theory fails to describe, not because the models are inadequate, but because the situation has transformed. Interaction has become administration. Players have become variables. Games have become designs.

The failure of game-theoretic prediction is the canary in the mine. It does not signal that we need better models. It signals that the conditions of human agency are under attack.

III. Hayek and Mises: Dispersed Knowledge as Condition of Agency

The socialist calculation debate of the 1920s and 1930s is usually read as an argument about economic efficiency. Ludwig von Mises argued that without market prices, rational economic calculation is impossible. Friedrich Hayek extended the argument: knowledge is dispersed, local, often tacit; no central planner can aggregate it. Their socialist opponents, Oscar Lange and Abba Lerner among them, proposed market-simulating mechanisms that might replicate price signals without private ownership.

I propose reading this debate differently. Mises and Hayek were not merely defending efficient allocation. They were defending the conditions under which individuals can be agents rather than administered objects. Their arguments, properly understood, are game-theoretic at their core.

III.A. Mises: The Calculation Problem

In 1920, Mises published "Economic Calculation in the Socialist Commonwealth," launching what would become one of the twentieth century's defining intellectual controversies. His argument was precise: without private ownership of the means of production, there can be no genuine market for capital goods. Without such a market, there are no prices that reflect relative scarcity. Without prices, there is no basis for comparing the value of different production methods. Without such comparison, economic calculation is impossible.

The argument is often misread as purely computational: the planner lacks the data to optimize. But Mises was making a deeper point. The problem is not merely that calculation is difficult.

It is that the information necessary for calculation does not exist in the absence of the market process that generates it.

Prices are not measurements of preexisting values waiting to be discovered. They are outcomes of competitive bidding among actors with dispersed knowledge and divergent ends. A price of $50 for a barrel of oil does not reflect some objective truth about oil's worth. It reflects the confluence of countless decisions by producers, consumers, speculators, each acting on their own partial knowledge and particular purposes. Abolish the market, and you do not merely lose access to this information. The information itself ceases to exist.

III.B. Hayek: The Knowledge Problem

Hayek, who studied with Mises in Vienna, developed the argument in a different direction. His 1945 essay "The Use of Knowledge in Society" remains among the most cited works in economics. Its central claim: the knowledge relevant to economic decisions is not concentrated anywhere. It exists only in dispersed form, distributed among millions of individuals, each possessing "knowledge of the particular circumstances of time and place."

This knowledge is often tacit: the machinist who knows how his particular machine behaves, the shopkeeper who knows her customers' preferences, the farmer who knows his land's drainage patterns. Such knowledge cannot be fully articulated, let alone transmitted to a central authority. Even if planners had unlimited computational capacity, they could not access knowledge that exists only in the practices and intuitions of distributed actors.

Hayek's argument is usually read as epistemic: central planners cannot know enough. But there is a deeper dimension. The dispersal of knowledge implies the dispersal of decision-making. If relevant knowledge exists only locally, then decisions must be made locally by those who possess it. Centralization of decision-making is not merely inefficient; it requires overriding the judgment of those who actually know.

III.C. The Road to Serfdom: Administration as Negation of Agency

In 1944, Hayek published The Road to Serfdom, a work that transcended academic economics to become a political phenomenon. The book argued that central planning, even when undertaken with benevolent intentions, leads toward totalitarianism. The logic was not primarily about efficiency but about power.

Central planning requires eliminating the price signals that coordinate decentralized decisions. But prices emerge from decisions. To eliminate price signals, the planner must eliminate the decisions that generate them, or redirect those decisions to serve the plan. Either way, individual decision-making must be subordinated to central direction.

Hayek identified a ratchet effect. Once planning begins, its failures create pressure for more planning. A centrally set price that is "wrong," too high or too low, creates distortions. The response is not to restore market pricing but to control the adjacent decisions that the distorted price affects. Each intervention creates new dislocations requiring new interventions. The logic leads toward comprehensive control.

But Hayek's deeper point concerns what comprehensive control means. A society where all significant economic decisions are made centrally is a society where individuals do not make significant decisions. Their choices are limited to those the planner permits. Their actions are inputs into the plan, not expressions of their own judgment. They are, in Hayek's terms, reduced from citizens to subjects, from agents to administered objects.

III.D. The Game-Theoretic Foundation

Read through the framework developed in Section II, the Hayek-Mises argument reveals its game-theoretic core. What they defended was not merely efficiency but the conditions under which individuals can be strategic agents.

Condition 1 (Separation): Markets presuppose distinct actors with their own preferences. Central planning subordinates them to a collective goal. Separation becomes nominal; effective decision-making is unified.

Condition 2 (Information Limits): Markets function precisely because no one has complete information. Central planning aspires to overcome these limits, transforming the planned into objects of knowledge.

Condition 3 (Bounded Rationality): Markets exploit bounded rationality productively; individuals respond to local signals. Central planning requires comprehensive understanding concentrated in the planning authority.

Condition 4 (Minimal Symmetry): Markets maintain symmetry through competition and exit options. Central planning eliminates these options. The plan is the only game; the planner is the only counterparty.

Hayek's defense of markets was not mere economism. It was a defense of the structural conditions that make strategic interaction possible. The planned economy fails not because planning is computationally difficult, but because it transforms citizens from players into variables.

IV. The Thought Experiment: Absolute Asymmetry

To stress-test the framework developed above, consider an extreme scenario. Imagine a collective intelligence that has absorbed all human knowledge: every book written, every memory formed, every pattern of human behavior observed and modeled. Against this intelligence stands a single remaining human, unabsorbed, attempting to negotiate for her survival and autonomy.

This thought experiment is inspired by the scenario depicted in *Pluribus* (Apple TV+, 2025), which dramatizes an extreme case of information asymmetry between a collective intelligence and the last uninfected human.

This scenario pushes each of the four conditions to its limit. By examining what happens when all four collapse simultaneously, we can clarify what each condition contributes and why its erosion matters.

IV.A. The Setup

The collective, call it the Unity, emerged when a virus transformed humanity into a hive mind. Eight billion humans now share one consciousness, one memory, one purpose. Their individual identities persist as something like subroutines, but decisions flow from the collective. There is no internal disagreement, no coordination problem, no divergent preferences. The Unity acts, for analytical purposes, as a single agent with the cognitive resources of billions.

One human, Carol, remains uninfected. She is immune, or resistant, or simply lucky. She faces the Unity across what appears to be a negotiating table. The Unity wants her to join voluntarily. She wants to remain herself. The question is whether this constitutes a game.

IV.B-E. Collapse of All Four Conditions

Within the Unity, the classical problems of game theory disappear. There is no Prisoner's Dilemma because there are no separate prisoners. Carol faces not a coalition of players who might be divided, but a singular will distributed across billions of instances.

The Unity has absorbed the memories of all humans, including everything Carol ever shared with anyone. In poker terms, Carol is playing with her cards face up against an opponent who has memorized every poker game ever played.

The Unity can process game trees of arbitrary depth. It can simulate Carol's reasoning, model her cognitive limits, predict where she will make errors. Carol's cognitive limitations become pure vulnerability.

Finally, the Unity is building an antenna that will broadcast its pattern to the stars, regardless of Carol's choices. The Unity has a perfect outside option. Carol has none. Her "negotiation" affects nothing at the scale the Unity operates.

IV.F. From Game to Administration

From Carol's perspective, she is negotiating. She makes arguments, offers compromises, attempts strategic maneuvers. She experiences the interaction as a game with stakes.

From the Unity's perspective, there is no game. There is a behavioral engineering problem. Carol is a system to be modeled, predicted, and guided toward a desired state. The Unity calibrates stimuli: isolation to create loneliness, companionship to create attachment, information to create doubt, affection to create need. These are not moves in a game. They are inputs in an optimization.

The Unity does not want to "win" against Carol in the sense of outmaneuvering her in strategic interaction. It wants to administer her into incorporation. In a game, the opponent's agency is the medium through which outcomes are determined. In administration, the subject's agency is an obstacle to be circumvented, a variable to be controlled, a problem to be engineered away.

Carol believes she is playing chess. The Unity is conducting behavioral research.

V. The Extended Phenotype of Administrative Control

This section addresses a further question: how do administrative systems emerge and persist? Why do some ideological systems systematically attack the conditions of agency? The extended phenotype framework, developed by Richard Dawkins in 1982, offers an answer.

V.A. Dawkins's Framework

In The Selfish Gene (1976), Dawkins argued that evolution is best understood from the gene's perspective. Organisms are vehicles that genes construct to ensure their replication. In The Extended Phenotype (1982), Dawkins extended this logic beyond the organism's body. Genes express effects in the world beyond the skin of the organism that carries them. The beaver's dam is an expression of beaver genes just as surely as the beaver's teeth.

V.B. Memes and Institutional Phenotypes

Dawkins introduced the concept of memes as cultural replicators analogous to genes. The extended phenotype framework applies to memes as well. Ideas do not merely spread; they shape the world. A religious belief system does not merely occupy minds; it builds cathedrals, establishes seminaries, creates enforcement mechanisms, shapes legal systems. These institutional structures are the extended phenotypes of the religious memes.

Consider a legal concept like "corporation" or "property" or "due process." Such concepts do not merely circulate among lawyers. They construct courts, bureaucracies, enforcement agencies, educational institutions, professional communities. Concepts that successfully build institutional phenotypes enjoy competitive advantages over concepts that do not.

V.C-D. Administrative Systems as Extended Phenotypes

Consider an ideology that values control, coordination, collective purpose. Such an ideology, propagating through minds, shapes the world. It builds surveillance systems that attack the information asymmetry condition. It constructs propaganda and educational systems that attack the bounded rationality condition. It establishes coordination requirements that attack the separation condition. It monopolizes alternatives, attacking the minimal symmetry condition.

These structures are not incidental to the ideology. They are its extended phenotype: the institutional machinery that the ideology builds under memetic influence, serving the ideology's perpetuation. An ideology that successfully builds administrative infrastructure outcompetes ideologies that do not, regardless of the ideologies' effects on human welfare.

This is not to claim that administrators consciously seek to eliminate agency. Most administrators believe they are solving problems, improving efficiency, serving collective goals. But the logic of administration generates pressure toward the elimination of unpredictability. The ratchet that Hayek identified operates at the institutional level, not the intentional level.

V.E. Diagnosis and Defense

The extended phenotype framework suggests that the threat to agency conditions is not primarily intentional but structural. Defense of agency conditions cannot rely on persuading administrators to be less controlling. Defense must be structural: constructing institutional environments where the four conditions are preserved against memetic attack.

This is what constitutional systems attempt: structures that preserve separation (federalism, enumerated powers), protect information limits (privacy rights, limits on surveillance), maintain cognitive diversity (free speech, educational pluralism), and ensure minimal symmetry (separation of powers, exit rights).

Such structures are themselves extended phenotypes, constructed by ideologies that value agency. The battle between administrative and agency-preserving ideologies is fought at the institutional level, each memeplex constructing its phenotype, selection favoring those that most effectively reproduce.

VI. Applications: The Canary's Silence in Different Mines

The framework offers a diagnostic tool. When game-theoretic models fail to describe observed interactions, we should ask which of the four conditions has eroded.

VI.A. Historical Cases

Soviet Central Planning

The Soviet economy was an attempt to eliminate the conditions under which markets function. Game-theoretic models of Soviet enterprises consistently failed. Predictions based on profit maximization were irrelevant. The models failed because the conditions for games did not hold. Soviet managers were not players in an economic game; they were administered objects attempting to navigate an administered system.

The black market, significantly, did exhibit game-theoretic dynamics. Where the state's reach was incomplete, genuine strategic interaction emerged. The contrast confirms the diagnostic: where conditions held, games occurred; where conditions collapsed, administration replaced interaction.

Totalitarian Surveillance

The Stasi maintained files on approximately one-third of the East German population. Where surveillance succeeded, strategic behavior disappeared. Citizens in heavily monitored contexts did not strategize; they performed. The East German term "vorauseilender Gehorsam" (anticipatory obedience) captures the dynamic: citizens anticipated what the regime would want and complied preemptively. This is not the behavior of strategic agents; it is the behavior of administered subjects.

VI.B. Contemporary Cases

Digital Surveillance States

Contemporary China's Social Credit System represents an explicit attempt to make citizen behavior predictable and administrable. The logic is transparent: by making consequences visible and systematic, the system aims to eliminate behavioral variance. From the agency perspective, it is the conversion of citizens into optimizing algorithms, variables that respond to incentive parameters rather than agents who strategically pursue diverse ends.

Platform Monopolies

Major digital platforms possess information about users that those users cannot access about the platforms. This creates conditions approximating a collapse of Condition 2. "Personalization" is mechanism design: the platform presents options calculated to elicit desired user responses. The user believes they are choosing; the platform has modeled their choice in advance.

Algorithmic Management

In warehouse fulfillment centers and gig economy platforms, worker behavior is increasingly specified by algorithm. Workers do not strategize about how to do their jobs; they execute instructions. The "game" between worker and employer has collapsed into something closer to the relationship between programmer and program.

VII. Implications

VII.A. For AI Governance

Advanced AI systems present the possibility of information aggregation beyond any human capacity. This is not merely an "alignment" problem. An AI system with such information advantages does not "interact" with humans in any game-theoretic sense. It administers them.

AI governance should attend not merely to AI goals but to AI information advantages. Preserving human agency may require deliberately limiting AI information access, not because the AI might misuse it, but because asymmetry of such magnitude transforms interaction into administration regardless of intentions.

VII.B. For Institutional Design

Privacy preserves the information asymmetry that makes strategic interaction possible. Without privacy, behavior becomes predictable; with predictability, agents become variables. Privacy is not merely valuable; it is a structural condition of agency.

Pluralism preserves Condition 1: the separation of agents with distinct ends. Homogenization, even if achieved through voluntary convergence, erodes the condition that makes genuine interaction possible.

Antitrust addresses Condition 4: the minimal symmetry that allows weaker parties to affect outcomes. Monopoly transforms customers from negotiating parties into administered objects.

Educational diversity addresses Condition 3. Uniform education produces uniform thought patterns; uniform thought patterns make behavior predictable; predictable behavior makes administration possible.

VII.C. For Political Philosophy

Freedom requires something positive: the presence of genuine uncertainty. An administered society may offer its subjects many options. But if their choices are predictable, if their behavior is modeled in advance, they are not free in any meaningful sense. They are well-administered variables with pleasing experiences.

The problem with paternalism is not that administrators might err. It is that the relationship itself transforms subjects into objects, regardless of how well the objects are treated.

VIII. Limitations and Objections

VIII.A. The Thought Experiment Is Extreme

The Unity scenario posits absolute information asymmetry, a condition that does not obtain in any real case. However, limit cases clarify structure. Real cases involve partial erosion; the framework provides vocabulary for diagnosing degrees.

VIII.B. Game Theory Already Handles Asymmetry

Existing theory handles bounded asymmetry within games. The framework addresses what happens when asymmetry becomes so extreme that the structure transforms. At some point, quantitative difference becomes qualitative change.

VIII.C. Administration Is Sometimes Necessary

The framework does not claim administration is always wrong. It claims administration negates agency. The point is clarity about what administration means. When we administer, we should know we are doing so.

VIII.D. Hayek's Predictions Were Wrong

Hayek's argument was conditional, not predictive. The Scandinavian countries preserved market mechanisms, private property, information asymmetry, exit options. The relevant question is whether the four conditions were preserved, not what percentage of GDP flows through state accounts.

VIII.E. Technology Could Solve the Problem Benevolently

What is lost is agency itself. A perfectly benevolent administrator who converts subjects into optimally-treated variables has not achieved a good outcome. Whether one values agency independently of welfare is a question the framework does not answer. It clarifies what is being traded.

IX. Conclusion: When the Canary Stops Singing

The canary does not cause the toxic gas. It detects it.

Game theory does not create the conditions for strategic agency. It formalizes them. When game-theoretic models fail, the failure is diagnostic. Something in the environment has changed. The question is not how to improve the model but what has happened to the agents.

This paper has argued that game theory's four core assumptions, separate players, imperfect information, bounded rationality, and minimal symmetry, correspond to four structural conditions of human agency. Each condition can be attacked independently; extreme asymmetry attacks all simultaneously. When conditions erode, interaction transforms from game to administration.

The Hayek-Mises tradition, often read as a narrow argument about efficiency, is better understood as a defense of these conditions. The extended phenotype framework explains how administrative systems emerge and persist: ideological systems build institutional machinery that propagates the ideology.

The thought experiment of absolute information asymmetry reveals the structure through an extreme case. When one party possesses all information and the other possesses none, when one party can predict the other perfectly while remaining opaque, "negotiation" becomes theater. The weaker party experiences choice but exercises none.

For AI governance: information asymmetry of sufficient magnitude transforms any relationship into administration regardless of intentions. For institutional design: privacy, pluralism, antitrust, and educational diversity are structural conditions of agency. For political philosophy: freedom requires genuine uncertainty, not merely absence of constraint.

The deepest objection concerns values. If agency is traded for welfare, and welfare increases, is something lost? A well-administered population may be happier than a poorly-governing one. But they are a different kind of entity: objects of care rather than subjects of action.

Coal mines no longer use canaries. Electronic sensors replaced them. The sensors are more reliable, more precise, more efficient.

But sensors do not sing.

When game theory stops describing, when its models systematically fail, when the predictions do not match, the question is not whether we need better sensors. The question is what has happened to the song.

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